

Python Practice Questions

Topics: Numeric, Boolean, String, List, Tuple, Mixed Data Types

Numeric

- Q1. Reverse a number without using string conversion.
- Q2. Check whether a number is an Armstrong number.
- Q3. Find the second largest digit in a number.
- Q4. Count even and odd digits.
- Q5. Find the product of digits except zero.
- Q6. Rotate digits left by one.
- Q7. Find the nearest multiple of 10.
- Q8. Check whether a number contains duplicate digits.

Boolean

- Q9. Predict the output of boolean expressions.
- Q10. Print 'Eligible' without using if when age $\geq$ 18 and salary $\geq$ 25000.
- Q11. Evaluate boolean expression result.
- Q12. Return True only if number divisible by 3 and not by 5.

String

- Q13. Count uppercase, lowercase, digits and special characters.
- Q14. Find the longest word.
- Q15. Remove duplicate characters while maintaining order.
- Q16. Check whether two strings are anagrams.
- Q17. Compress a string (e.g., aaabb -> a3b2).
- Q18. Print the first non-repeating character.
- Q19. Find all vowels with their positions.
- Q20. Reverse words only.

List

- Q21. Find the second largest element without sort().
- Q22. Move all zeros to the end.
- Q23. Find duplicate elements.
- Q24. Split list into even-index and odd-index lists.
- Q25. Rotate list right by 2 positions.
- Q26. Flatten a one-level nested list.
- Q27. Find the missing number.
- Q28. Replace every even number with its square.
- Q29. Find frequency of each element without count().
- Q30. Find maximum difference between adjacent elements.

Tuple

- Q31. Find the maximum element.
- Q32. Convert nested tuple into a single tuple.
- Q33. Find common elements of two tuples.
- Q34. Reverse a tuple without slicing.
- Q35. Count positive and negative numbers.
- Q36. Remove duplicate values while keeping tuple output.
- Q37. Find the pair having maximum sum.
- Q38. Swap first and last elements.

Mixed Data Types

- Q39. Count each datatype in a mixed list.
- Q40. Extract only numeric values.
- Q41. Find the longest string in a mixed list.
- Q42. Replace every string with its length.
- Q43. Find the sum of all numeric values in a nested list.
- Q44. Count mutable and immutable objects.

Q45. Separate elements according to datatype.

Challenge

Q46. Find the second largest unique number in a nested list.

Q47. Rotate a string without using slicing.

Q48. Check whether two lists are rotations of each other.

Q49. Calculate the sum of numeric values at even indices only.

Q50. Replace every prime number in a nested list with its square.

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